

DIGITAL IMAGE PROCESSING

MINI PROJECT REPORT

Plant Leaf Disease Region Segmentation

Using Classical Digital Image Processing Techniques

Bachelor of Computer Applications

Digital Image Processing

Submitted by:

Nakshathra D

Register No.: 24UBC146

Academic Year: 2026

1. Problem Statement

Plant leaf diseases often appear as localized spots or affected regions on the leaf surface. Identifying these regions from leaf images is useful for visual disease assessment and can support automated plant-health analysis. This project segments diseased or affected regions using classical Digital Image Processing techniques and compares the results with available ground-truth masks.

2. Objectives

- To load and inspect plant leaf images together with their ground-truth disease masks.
- To preprocess images using grayscale conversion and Gaussian smoothing.
- To apply Otsu and Adaptive thresholding as intensity-based segmentation methods.
- To apply HSV color-based segmentation to identify yellow/brown affected regions.
- To improve segmentation masks using morphological opening and closing.
- To evaluate methods using IoU, Dice Coefficient and Pixel Accuracy.
- To compare the methods on 100 matched images and analyze their strengths and limitations.

3. Dataset Description

The project uses the Kaggle dataset “Plants Leaves Dataset with Masks for Segmentation”. The downloaded dataset contains separate images and masks directories inside segData.

Property	Value
Number of images	1611
Number of masks	1611
Matching pairs	1611
Image size	128 × 128 pixels
Image type	RGB
Raw mask values	0 and 1
Ground-truth meaning	0 = background/healthy tissue; 1 = diseased/affected region
Evaluation subset	100 matched images

The raw masks are converted to an 8-bit binary representation: 0 for background and 255 for the diseased/affected region.

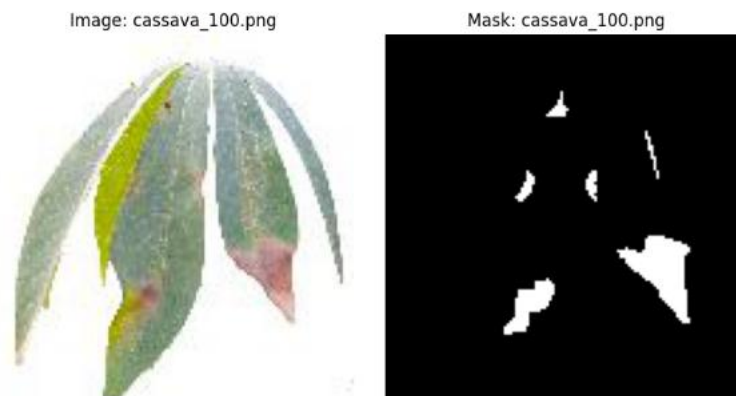


Figure 1: Sample plant leaf image and corresponding ground-truth mask

4. Methodology

1. Dataset extraction and image-mask matching
2. Sample image and ground-truth mask inspection
3. Ground-truth mask preparation
4. Grayscale conversion and Gaussian smoothing
5. Otsu thresholding
6. Adaptive thresholding
7. HSV color-based segmentation
8. Morphological opening and closing
9. Quantitative evaluation using IoU, Dice and Pixel Accuracy
10. Evaluation across 100 images and comparison of average results

5. Image Preprocessing

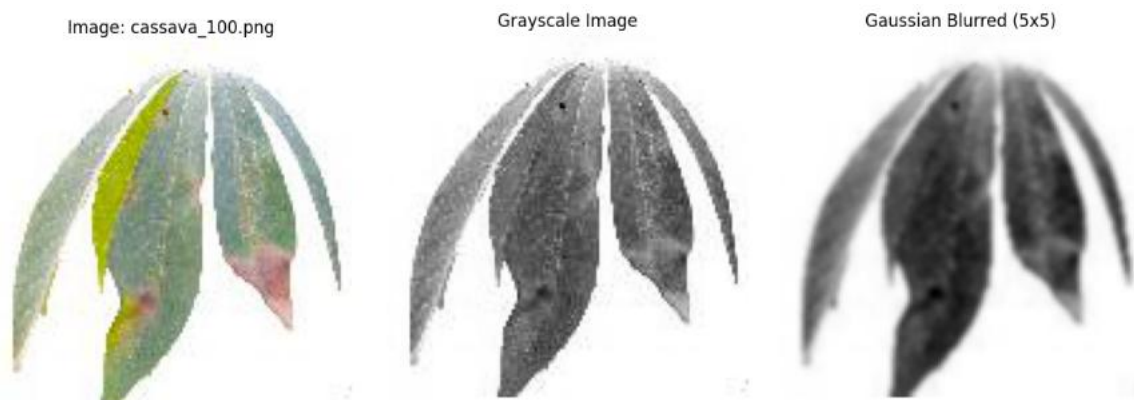


Figure 2: Image preprocessing using grayscale conversion and Gaussian smoothing

5.1 Grayscale Conversion

The RGB image is converted to grayscale using OpenCV, producing a single intensity channel for intensity-based thresholding.

5.2 Gaussian Smoothing

A 5×5 Gaussian filter is applied to reduce small noise and minor texture variations before Otsu and Adaptive thresholding.

6. Segmentation Techniques

6.1 Otsu Thresholding

Otsu thresholding is used as a baseline global thresholding method. It automatically selects one threshold from the image histogram. For the sample image, the calculated threshold was 204.0.

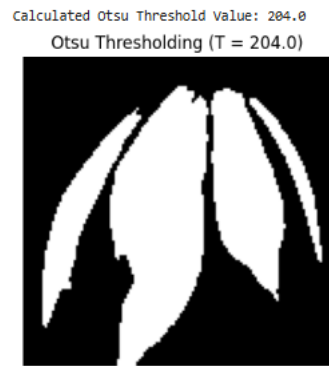


Figure 3: Disease-region segmentation using Otsu thresholding

6.2 Adaptive Thresholding

Adaptive Gaussian thresholding uses local neighbourhood information instead of one global threshold. The implementation uses a neighbourhood size of 11×11 and $C = 2$.

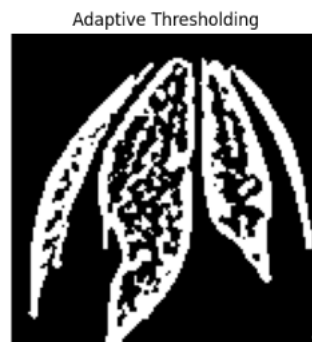


Figure 4: Segmentation using Adaptive Gaussian Thresholding

6.3 HSV Color-Based Segmentation

The RGB image is converted to HSV. A selected HSV range is used to identify potential yellow/brown disease-colored regions. A leaf-region mask is also used to reduce detection of the light background.

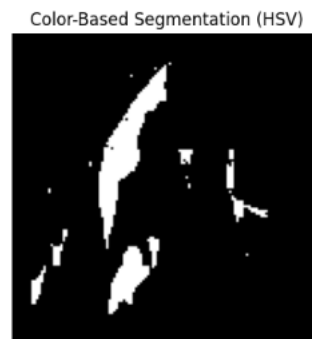


Figure 5: HSV color-based segmentation of affected regions

6.4 Morphological Processing

A 3×3 kernel is used. Opening removes small isolated noise, while closing fills small holes and gaps in detected regions.

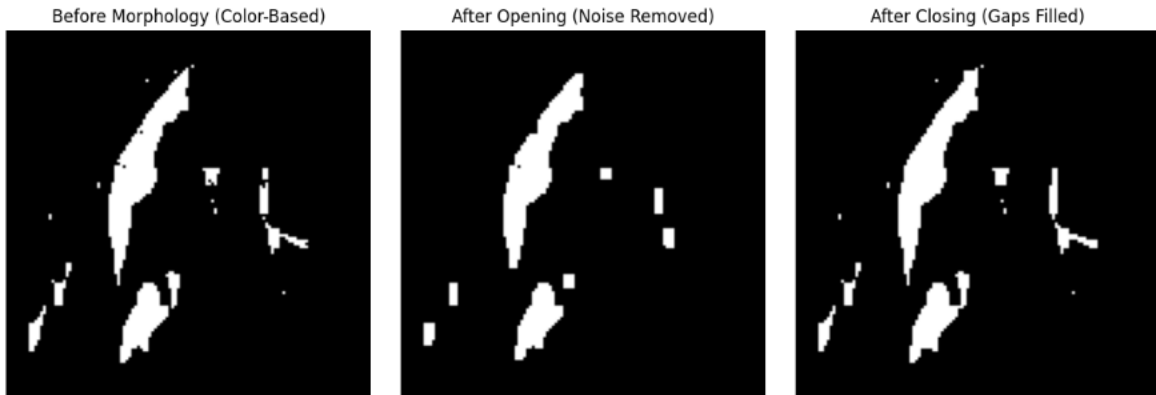


Figure 6: Effect of morphological opening and closing on the color-based segmentation mask

7. Implementation

The project was implemented in Python using Google Colab. OpenCV was used for image reading, color-space conversion, filtering, thresholding and morphological operations. NumPy was used for array and mask processing, while Pandas was used to store and summarize the evaluation results. Matplotlib was used to display the processed images and comparison graphs.

The implementation first loads the plant leaf images and their corresponding ground-truth masks. The images are preprocessed using grayscale conversion and Gaussian smoothing. Otsu and Adaptive thresholding are then applied as intensity-based methods, while HSV color-based segmentation is used to identify potential diseased regions based on their color characteristics. Morphological opening and closing are applied to improve the resulting masks.

Each predicted mask is compared with its corresponding ground-truth mask using IoU, Dice Coefficient and Pixel Accuracy. The complete experiment is performed on 100 matched image-mask pairs, and the average values are used to compare the segmentation methods.

8. Evaluation Metrics

8.1 Intersection over Union (IoU)

IoU measures overlap between the predicted and ground-truth regions: intersection divided by union.

8.2 Dice Coefficient

Dice measures overlap using twice the intersection divided by the total number of positive pixels in both masks.

8.3 Pixel Accuracy

Pixel Accuracy is the proportion of correctly classified pixels. Because disease regions occupy a relatively small part of many images, Pixel Accuracy can be high even when disease-region overlap is limited.

9. Experimental Results

The methods were evaluated on 100 matched images. Each method was compared with its corresponding ground-truth mask and the metrics were averaged across the 100 images.

Method	Average IoU	Average Dice	Average Pixel Accuracy
Adaptive Thresholding	0.1001	0.1590	0.8176
Color-Based	0.2731	0.3886	0.9350
Color-Based + Closing	0.2900	0.4053	0.9341
Color-Based + Opening	0.2443	0.3481	0.9390
Otsu Thresholding	0.1669	0.2160	0.7542

Color-Based + Closing obtained the highest average IoU (0.2900) and Dice (0.4053). Color-Based + Opening obtained the highest Pixel Accuracy (0.9390).

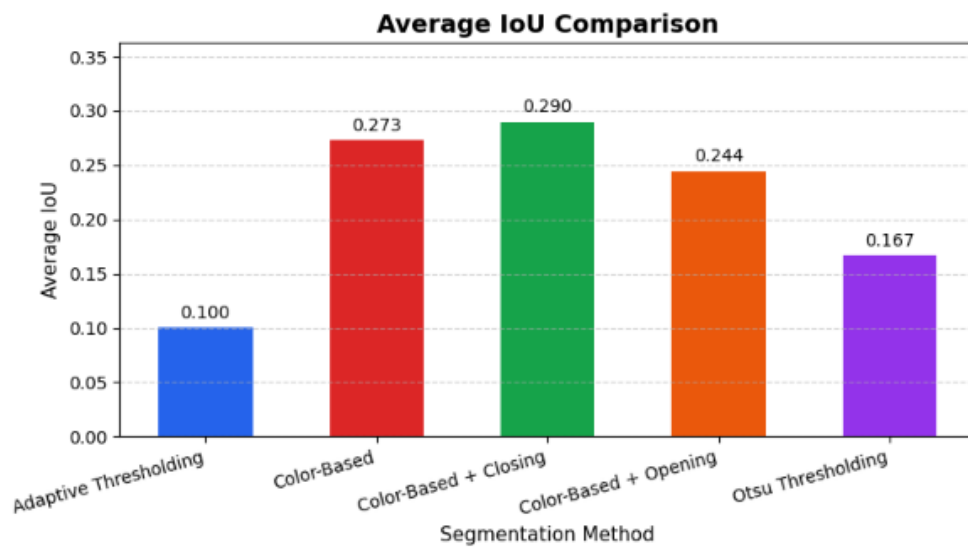


Figure 7: Average IoU Comparison

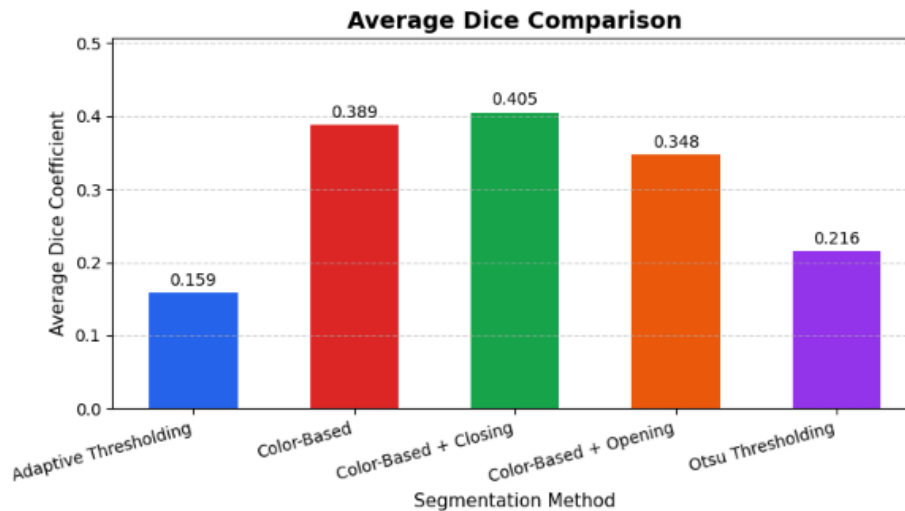


Figure 8: Average Dice Comparison

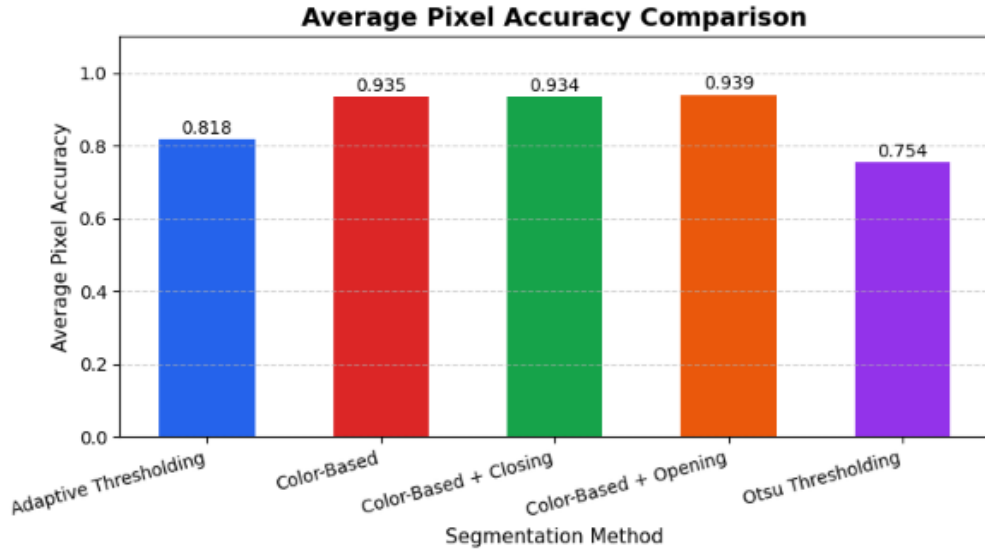


Figure 9: Average Pixel Accuracy Comparison

10. Result Analysis

The grayscale thresholding methods were less effective at isolating localized disease regions because leaf veins, shadows and edges can have intensity values similar to affected regions. HSV-based segmentation uses color information and produced higher average overlap than the grayscale baselines in this experiment.

Morphological closing slightly improved average IoU and Dice over the unprocessed color-based mask, while opening reduced small noise but produced lower IoU and Dice than closing in the 100-image experiment.

The visual results also show false-positive regions in some images. Healthy leaf areas with colors similar to the selected disease range can therefore be detected as disease regions. The project is a comparison of classical segmentation techniques, not a complete disease-diagnosis system.

11. Visual Comparison of Results

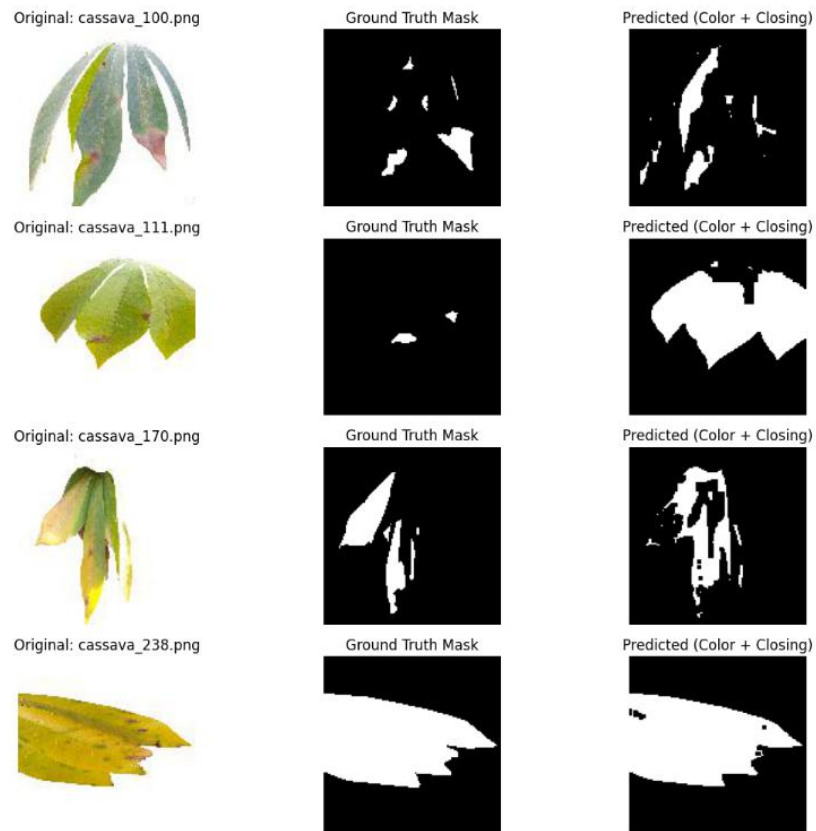


Figure 10: Visual comparison between original images, ground-truth masks and predicted disease-region masks using Color-Based + Closing

12. Limitations

- Fixed HSV threshold ranges may not work equally well under different lighting conditions.
- Different plant species and disease types can have different lesion colors.
- Healthy leaf pigmentation can sometimes be similar to the selected disease-color range.
- Pixel Accuracy can be misleading when disease regions occupy only a small portion of the image.
- The experiment uses 100 images as an evaluation subset rather than the complete dataset.

13. Future Scope

- Use adaptive color clustering methods such as K-Means in alternative color spaces.
- Use texture features such as GLCM or LBP to distinguish disease regions from natural leaf structures.
- Explore more advanced segmentation methods for greater robustness across species and lighting conditions.
- Evaluate the approach on a larger and more diverse image set.

14. Conclusion

This project implemented a classical Digital Image Processing approach for plant leaf disease-region segmentation. The workflow included preprocessing, Otsu thresholding, Adaptive thresholding, HSV color-based segmentation and morphological processing. The methods were quantitatively compared against ground-truth masks using IoU, Dice Coefficient and Pixel Accuracy.

Across the 100-image evaluation subset, Color-Based + Closing achieved the highest average IoU and Dice values, while Color-Based + Opening achieved the highest Pixel Accuracy. The experiment shows that color information was useful for this dataset, while also demonstrating the limitations of fixed classical thresholds.

15. Tools and Technologies Used

- Python
- Google Colab
- OpenCV
- NumPy
- Pandas
- Matplotlib